QUICKSTEP COMPUTER CENTER

National Accreditation Board of Education Training. (NABET)- Quality council of India) An ISO 9001:2008

Data Science with AI - ML

Module1:Introduction to Data Science

- Selecting rows/observations
- Rounding Number
- Selecting columns/fields
- Merging data
- Dataaggregation
  Datamungingtechniques

Module2:Introduction to Python

- What is Python?
- Why Python?
- Installing Python
- Python IDEs
- Jupyter Notebook Overview

Module3:Python Basics

- Python Basic Data types
- Lists
- Slicing
- IF statements
- Loops
- Dictionaries
- Tuples
- Functions
- Array
- Selection by position & Labels

Module4:Python Packages

- Pandas
- Numpy
- Scipy
- Mat-plotlibrary

Module5:Pandasmodule

- Pandas series
- Pandas data frame
- Reading CSV files
- Saving in Python data
- Loading Python data objects
- Cleaning the data

Module6:Numpy Module

- Creating numpy
- Array indexing
- Array slicing
- Shape and reshape
- Copy and review
- Array join
- Sorting
- Seaborn

Module7:Manipulating Data

- Selecting rows/observations
- Rounding Number
- Selecting columns/fields
- Merging data
- Data aggregation
- Data munging techniques

Module8:Scipy Module

- Scipy constant
- Scipy optimizer
- Scipy graph
- Scipy matlab array

Module9:Matplotlib Module

- Matplotlib pyplot
- Matplotlib plotting
- Matplotlib marker
- Matplotlib line
- Matplotlib grid
- Matplotlib scatter
- Matplotlib bar
- Matplotlib pie

Module11:Statistics Basics

- Central Tendency
  - Mean
  - Median
  - Mode
  - Skewness
  - Normal Distribution
- Probability Basics
  - What does it mean by probability?
  - Types of Probability
  - ODDS Ratio?
- Standard Deviation
  - Data deviation & distribution
  - Variance
- Biasvariance Trade off
  - Under fitting
  - Over fitting
- Distance metrics
  - Euclidean Distance
  - Manhattan Distance
- Outlier analysis
  - What is an Outlier?
  - Inter Quartile Range
  - Box & whisker plot
  - Upper Whisker
  - Lower Whisker
  - Scatter plot

    - Cook’s Distance
- Missing Value treatment
    - What is NA?
    - Central Imputation
    - KNN imputation
    - Dummification
- Correlation
    - Pearson correlation
    - Positive & Negative correlation

Module8: Error Metrics

- Classification
    - Confusion Matrix
    - Precision
    - Recall
    - Specificity
    - F1Score
- Regression
    - MSE
    - RMSE
    - MAPE

Machine Learning

Module1:SupervisedLearning

- Linear Regression
    - Linear Equation
    - Slope
    - Intercept
    - Rsquare value
- Logistic regression
    - ODD Sratio
    - Probability of success
    - Probability of failure Bias Variance Tradeoff
    - ROC curve
    - Bias Variance Tradeoff

Module2:Unsupervised Learning

- K-Means
- K-Means++
- Hierarchical Clustering

Module3:SVM

- Support Vectors
- Hyper planes
- 2-DCase
- Linear Hyperplane

Module4:SVMKernal

- Linear

- Radial
- polynomial

Module5:Other Machine Learning Algorithms

- K–Nearest Neighbour
- Naïve Bayes Classifier
- Decision Tree–CART
- DecisionTree–C50
- Random Forest

ArtificialIntelligeceModule1:AI Introduction

- Perceptron
- Multi-Layer perceptron
- Markov Decision Process
- Logical Agent & First Order Logic
- AL Applications

Deep Learning

Module1:Deep Learning Algorithms

- CNN –Convolutional Neural Network

- RNN–Recurrent Neural Network
- ANN–Artificial Neural Network

Module2:Introductionto NLP

- Text Pre-processing
- Noise Removal
- Lexicon Normalization
- Lemmatization
- Stemming
- Object Standardization

Module3:Text to Features (Feature Engineering)

- Syntactical Parsing
- Dependency Grammar
- Part of Speech Tagging
- Entity Parsing
- Named Entity Recognition
- Topic Modeling
- N-Grams
- TF–IDF
- Frequency/Density Features
- Word Embedding's

Module4:Tasks of NLP

- Text Classification
- Text Matching
- Levenshte in Distance
- Phonetic Matching
- Flexible String Matching